

Number-Systems-and-Number-Sense

CSAT semantic master • csat-02 • g1 • approved: false

01

Number universe

Natural, whole, integer, rational, irrational and real numbers.

02

Divisibility

Use prime-factor and digit tests only within their valid base-ten conditions.

03

Primes and factors

Factorise before counting divisors or comparing powers.

04

HCF and LCM

Use gcd-lcm relations with integer and positivity checks.

05

Remainders

Reduce early and preserve the modulus.

06

Unit digits

Use cyclicity and treat exponent-zero cases separately.

07

Digits and place value

Translate reversal and digit count into base-ten equations.

08

Powers and factorials

Use repeated division for prime exponents and trailing zeros.

09

Fractions and decimals

Convert recurring forms through algebra, not memorised guesses.

10

Series

Test differences, ratios, alternation and position rules.

11

Magnitude and estimation

Bound before calculating exactly.

12

Verification

Plug back, check parity, digit count, remainder and order of magnitude.